# Implementing Dataloaders

| Index | label | Text |
|---|---|---|
| 0 | ham | -- |
| 1 | ham | . - - - |
| 2 | spam | . - - |
| 3 | ham | - - - |
| 4 | ham | . . . . |

→ Text messages are of varying length
↓
To batch these messages, we have 2 options

(1) Truncate all messages to length of shortest message
(2) Pad all messages to length of longest message

BPE tokenizer

input text → 1) Tokenize texts → token IDs → 2) Pad to longest sequence → padded token IDs (endoftext token)

"First message" → 717, 2410 → 717, 2410, 50256, 50256, 50256, ..., 50256 ← we use 50256 as padding token

"another text message" → 1194, 2420, 3275 → 1194, 2420, 3275, 50256, ..., 50256

"<|endoftext|>": 50256

"This is the third text message which is very long" → 1212, 318, 262, 2368, 2420, 3275, 11, 543, 318, 845, 890 → same as before ← no padding in this last example b/c it is longest message

Procedure!

Input text → Tokenize → Ensure that all sentences are same length

| Original message text | Padded to 120 tokens: 1, 2, 3, ..., 120 | Class label/array where 1 stands for "spam" and 0 for "ham" (not spam) → target |
|---|---|---|
| At what time you... (1) | 2953, 644, 640, ..., 50256 | 0 |
| XMAS Prize draws!... (2) | 53, 31180, 15895, ..., 50256 | 1 |
| Dear voucher holder (3) | 20266, 4062, 1562, ..., 50256 | 1 |
| ... | ... ... ... | ... |
| K, I will send again (8) | 42, 13, 314, ..., 50256 | 0 |
| | 8x120 | 8x1 |

Training Batch of 8 samples

Each batch consists of 8 training examples

Each entry (row) represents the token IDs corresponding to the original message text

Class label of $8^{th}$ training example

747 "spam"
747 "no spam"
1494 total samples
↳ 70% in training = 1045

[8, 120] → [130, 8, 120] → train_loader dimensions

[38, 8, 120] → test_loader dims.

[19, 8, 120] → val_loader dims.

8 samples each batch → # of batches $= \frac{1045}{8} \approx 130$ total training batches